Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Lindner, Jonathan R M.D.		
Project Title:	Biomarker Assessment for Identifying Heightened Risk for Cardiovascular Complications During Long-duration Space Missions		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	97239-3011	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	06/01/2015	**End Date:**	05/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Change in period of performance to 6/1/2015-5/31/2017 (originally 5/1/15-4/30/17), per NSBRI (Ed., 6/29/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fazio, Sergio M.D., Ph.D. (Oregon Health & Science University) Spellman, Paul Ph.D. (Oregon Health & Science University) Stevens, Jan Ph.D. (Oregon Health & Science University) Raber, Jacob Ph.D. (Oregon Health & Science University) Turker, Mitchell Ph.D. (Oregon Health & Science University) Ferencik, Maros M.D. (Oregon Health & Science University)		
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Task Description:

Projects Aims/Objectives: The goals for atherosclerotic screening of astronauts in preparation for space exploration are quite different than that used for screening of the general population. Since an in-flight atherosclerotic complications would put the individual and the entire mission at great peril, any screening algorithm is required to definitively identify the predisposition to future atherothrombic events irrespective of the presence of significant stenosis. The aim of this proposal is to use state-of-the-art omics approaches for improving atherosclerotic risk assessment in apparently healthy individuals. The research program was divided into two main aims; first to identify representative individuals with various risk for cardiovascular outcomes and second to use personal metabolomics and lipidomics profiling to correlate risk of disease with molecular biomarkers. Risk was stratified in a two-step process, first involving the recruitment of asymptomatic subjects with either very low, moderate, or very high risk features on coronary CT angiography features. A second evaluation method (myocardial contrast echocardiography) was then applied to assess endothelial function within each cohort since the vascular endothelium is a key target in the pathophysiology of accelerated atherosclerotic disease during deep space exploration. For each risk category group, we performed integrative omic analysis from serum. In addition, whole genomic sequencing was performed on peripheral blood mononuclear cells (PBMCs) in all individuals (initially we had planned to study whole genome sequencing (WGS) only in 20 subjects but costs for these tests decreased in the 2-year course of the study to allow WGS in all subjects). The purpose of the study was to identify any omics patterns that can be used as a strong predictor of risk that may be particularly useful for identifying individuals with heightened risk for the effects of prolonged space exploration. These tests may not only be useful for initial screening, but also for periodic evaluation during mission lead time and also for investigational assessment of the effects of space radiation on the International Space Station (ISS) or during space exploration missions. To perform these studies, we assembled a multidisciplinary team of investigators with expertise in non-invasive cardiovascular imaging in risk assessment, metabolomics, proteomics, endothelial biology, and bioinformatics/biostatistics.
Key findings: Enrollment and all study protocols have been completed in all subjects which has yielded a tremendous amount of high dimensional omics data that now must be registered according to imaging stratification and adjusted for clinical variables by implementing two machine learning methods to build and compare prediction models, including Elastic Net regression and Random Forest classification. This analysis is still ongoing. However, a summary of key findings are: 1. We have defined that half of all astronaut-aged individuals referred for CT-A for symptoms but without significant coronary artery disease (CAD) have evidence of microvascular dysfunction; half of whom have severe microvascular dysfunction. Also, not all subjects with high risk CAD had significant coronary calcium. 2. Brachial artery flow-mediated dilation (FMD), which has been hypothesized as a method for assessing endothelial function and dysfunction, is a poor indicator of coronary microvascular dysfunction. 3. Unique lipidomic features have been identified which through hierarchical clustering analysis show an extremely high predictive value for either non-obstructive but high risk coronary artery disease, or microvascular dysfunction. There is some overlap of features for each process, including oxidative pathways involving lyso-PC. 4. Plasma metabolomics showed a high degree of discrimination for CAD; in particular high deoxycarnitine and lower levels of its active metabolite carnitine was a strong predictor implicating pathways that are essential essential for LV function through fatty acid ß-oxidation, and regulation of blood pressure and perfusion, and atherosclerotic disease via adiponectin-stimulated production of nitric oxide or gut microbiome production of TMAO.

Impact: There are several important insights provided by the results of our studies. We have identified new and potentially important imaging protocols that can be used to best assess presence of subclinical atherosclerosis and coronary microvascular dysfunction. We have also demonstrated that some tests that are conventionally used (FMD) lack predictive power. We have also identified unique lipidomic and metabolomics features, many of which are related to oxidative stress and carnitine metabolism which are known targets of space radiation, that have strong predictive power for high risk but non-obstructive CAD and/or microvascular dysfunction. These omic pathways can be tested prospectively for their utility in either predicting cardiovascular or central nervous system (CNS) events in humans or for assessing pathophysiology and mitigation strategies for adverse effects of space radiation on the cardiovascular system in pre-clinical models.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The Earth-based impacts and innovative technologies resulting from the research program are below:
1. Data from this National Space Biomedical Research Institute (NSBRI) project have provided strong evidence the brachial flow-mediated dilation approaches for assessing endothelial dysfunction are not suitable for identifying coronary microvascular dysfunction.

2. Data from our study also indicate that the non-invasive assessment for coronary microvascular dysfunction with regadenoson vasodilator stress myocardial contrast echocardiography could be a valuable tool in assessing patients who have ischemic symptoms but no obstructive CAD. The technique is much more feasible to use than cath-lab based coronary assessment since it is non-invasive, safe, inexpensive, and requires only 15 min to perform.

3. Several lipidomics and metabolomics pathways that involve increased oxidative stress, nitric oxide production, and possibly gut microbiome effects through TMAO (trimethylamine-N-oxide) have been identified with regards to early atherosclerotic disease and could be useful as plasma biomarkers for atherosclerotic risk. Importantly for NASA, some of these same lipidomic and metabolomics pathway changes (lyso-phosphatidylcholine and deoxycarnitine) have also been found by our group in mice exposed to 56Fe irradiation, indicating that they may be useful in risk prediction algorithms for astronauts considered for deep space missions.

4. Our results are the first description of integrative omics in microvascular dysfunction; some of the patterns appear to have overlap with atherosclerotic risk pathways.

Task Progress:

(1) Subject recruitment has ended and all subject protocols have been completed
(2) All myocardial contrast echocardiography microvascular perfusion and CT coronary angio data have been analyzed

(3) Metabolomic and lipidomic analysis has been completed

(4) Whole genome sequencing has been completed

(5) Database population has been completed

(6) Data analysis including machine based learning algorithms are 80% complete

	(7) Two separate manuscripts are in preparation.
Bibliography Type:	Description: (Last Updated: 05/26/2021)
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